

AKSHAY CHOUDHARY

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Education

National Institute of Technology, Rourkela

Bachelor of Technology in Biotechnology — CGPA: 8.59

Aug 2023 – Present

Rourkela, Odisha

Experience

Research Intern

May 2026 – Present

IIT Kharagpur (Grisha Research Internship Program)

- Designing and evaluating deep learning experiments involving representation learning, model benchmarking, and statistical performance analysis.
- Reproduced and evaluated research-grade neural architectures using PyTorch and scientific datasets.
- Designed reproducible experimentation pipelines for model training, hyperparameter tuning, and performance evaluation.
- Investigating representation learning techniques for improving predictive performance in scientific machine learning tasks.

Open Source Contributions

PyCQA/Pylint Contributor | *Python, Static Analysis, Open Source*

GitHub

- * Contributed fixes to a widely used Python static analysis tool improving linting accuracy and Python compatibility.
- * Merged multiple pull requests addressing static analysis correctness, inference handling, and Python compatibility issues.
- * Investigated and resolved edge cases involving type inference, AST analysis, and static code checking.
- * Collaborated with maintainers through code reviews, issue discussions, and iterative patch development.

Projects

MiniML - Machine Learning Library | *Python, NumPy, PyPI, GitHub, CI/CD*

GitHub

- * Developed and published a machine learning library on PyPI implementing 9 machine learning algorithms and 15+ unit tests with automated CI/CD.
- * Designed Scikit-learn-inspired APIs supporting model training, preprocessing, evaluation, and inference workflows.
- * Built modular and extensible architecture with automated testing, package versioning, and CI/CD through GitHub Actions.

AI-Powered Meeting Minutes Generator | *Pegasus, FastAPI, Hugging Face, React, Vercel*

GitHub

- * Built an end-to-end LLM-powered system for automatic meeting transcription and abstractive summarization.
- * Developed data preprocessing pipelines for transcript cleaning, tokenization, and conversational normalization.
- * Fine-tuned and evaluated Pegasus-based summarization workflows, improving summary quality measured through ROUGE evaluation.
- * Designed scalable inference APIs and integrated frontend workflows for real-time summary generation.

Multithreaded HTTP Server in C | *C, POSIX Sockets, Multithreading, Linux*

GitHub

- * Built a multithreaded HTTP server in C using POSIX sockets and pthreads for concurrent request handling.
- * Implemented thread-pool scheduling using mutexes and condition variables.
- * Designed HTTP parsing and concurrent connection handling while strengthening OS and networking fundamentals.

Technical Skills

Programming: Python, SQL, C++, C, JavaScript

Machine Learning & Deep Learning: PyTorch, Deep Learning, Scikit-learn, Hugging Face, NLP

Backend: FastAPI, REST APIs, Flask

Data Science: Pandas, NumPy, Statistics, Matplotlib, Feature Engineering, Model Evaluation, Cross Validation, EDA, Visualization

Vector Databases: FAISS, Pinecone, ChromaDB

Developer Tools: Git, GitHub, Docker, Jupyter Notebook, Streamlit

Achievements

- Top 10 Finalist – Teach-a-thon Hackathon (500+ teams)
- Codeforces (Max Rating: 922) and CodeChef 3-Star (Max Rating: 1346)
- Solved 300+ DSA problems
- Organized HackOdisha 4.0, a 36-hour national hackathon involving 100+ participating teams.